

Revised Generation Interconnection System Impact Study Report

For Queue Project Q18-1059

Scenic Hills 138 kV
January 31, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1059. The Interconnection Customer proposes a 211.9 MW solar photovoltaic facility interconnecting at the Scenic Hills 138 kV substation in the Pennsylvania area. The request was received on September 08, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 375 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$82,118,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Vera-Norgate 138kV Line	Carbon Ctr-Caldwell 345kV	154.2	183	16.9
Pixley-Red Granite 138kV	Piney Crk Project-Caldwell 345kV line	118.9	205	5.0
Mt. Rose-Amherst Substation 138kV	Carbon Ctr 500-345 kV Auto	112.3	144	18.0
Caldwell-Red Granite 138kV Line	Caldwell-Piney Creek Project Substation 345kV line	110.2	217	5.0
Kathywood Bamboo-Mid America 138kV	Orchard Rd Substation-Shawvill Station 500kV L	109.2	198	5.4
Amherst-Vfw Park 138kV L	Carbon Center 500-345 kV Transformer	108.9	209	12.4
Anniston-Mann 138kV line	Caldwell-Carbon Ctr 345kV	103.5	172	7.2
Moshannon-Ltg-Seagrav Substation 138kV	Shawville Sta-Orchard Road 500kV	101.4	244	6.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Norgate-Vera 138kV	Rebuild line with higher-capacity conductor	\$28,071,000	100% (\$28,071,000)
Pixley-Red Granite 138kV	Rebuild line with higher-capacity conductor	\$13,756,000	100% (\$13,756,000)
Mt. Rose-Amherst Substation 138kV	Rebuild line with higher-capacity conductor	\$5,971,000	100% (\$5,971,000)
Caldwell-Red Granite 138kV	Rebuild line with higher-capacity conductor	\$21,536,000	100% (\$21,536,000)
Kathywood Bamboo-Mid America 138kV	Reconductor line	\$2,363,000	28% (\$662,000)
Amherst-Vfw Park 138kV Line	Reconductor line	\$2,548,000	100% (\$2,548,000)
Mann-Anniston 138kV	Reconductor line	\$8,255,000	100% (\$8,255,000)
Ltg-Seagrav-Moshannon 138kV	Reconductor line	\$2,185,000	60% (\$1,319,000)
	Total allocated to project		\$82,118,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.